

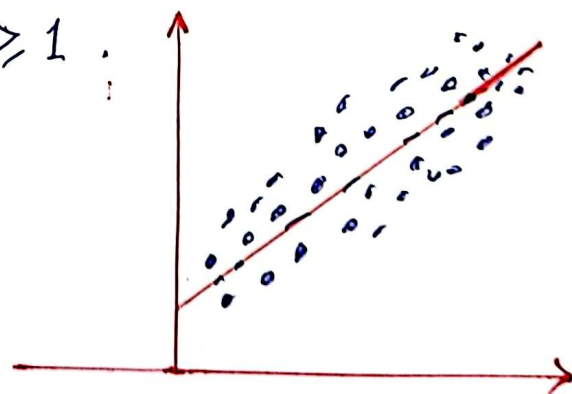
Curve Fitting :

①

Curve fitting is to form an equation of the curve from a given data. There are two methods for fitting curve: ① Graphical method ② Method of least square.

Graphical Method :

graphical method is simple here we have to draw a curve passing through the maximum number of points. In practice it's difficult for forming of the points. The simple curve can be fitted to number of points is line for which $n \geq 1$.



Method of least square :

Let $x_1, x_2, \dots, x_n$ and

$y_1, y_2, \dots, y_n$, be n sets of observations

(data) of the selected data, and $y = a + bx$ be the straight line to be fitted.

(2)

Let $y = f(x)$ be the solution suggested between x & y .
The difference between the expected value and observed value when $x = x_i$ the observed values is y_i , the expected value is $f(x_i)$ error is :

$$e_i = y_i - f(x_i)$$

Our goal is to find the functional relationship between the observed values and expected values. The difference between the observed and expected values is known as residual (error).

We want to minimize these residual (error), since these difference may be positive in some cases and of squares of these residual a minimum. This method is called the method of least square.

How to derive the normal equations of straight line?

Let the straight line be $f(x) = y = a + bx$ --- (1)

Then let the straight line in (1) passes through the data points: $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$. That is:

$$y_i = a + bx_i \text{ --- (2)}$$

(3)

The error e_i between the observed and expected value of $y = y_i$ is defined as:

$$e_i = y_i - f(x_i), \quad i = 1, 2, \dots, n.$$

Then $e_i = y_i - (a + bx_i)$ (we used equ (2))

Sum of the squares of these errors is:

$$\sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - (a + bx_i))^2. \quad \text{let } S = \sum_{i=1}^n e_i^2,$$

then, $S = \sum_{i=1}^n (y_i - (a + bx_i))^2 = \text{least square error.}$

Now, for S to be minimum, we do

$$\frac{\partial S}{\partial a} = 0, \quad \text{and} \quad \frac{\partial S}{\partial b} = 0, \quad \text{so,}$$

$$\frac{\partial S}{\partial a} = 0 = 2 \sum_{i=1}^n (y_i - (a + bx_i))(-1) = 0$$

$$\Rightarrow \sum_{i=1}^n y_i - \sum_{i=1}^n a - \sum_{i=1}^n bx_i = 0$$

$$\therefore \boxed{\sum_{i=1}^n y_i = n \cdot a + b \sum_{i=1}^n x_i} \quad \dots (1)$$

and,

$$\frac{\partial S}{\partial b} = 0 = 2 \sum_{i=1}^n (y_i - a - bx_i)(-x_i) = 0$$

$$\sum_{i=1}^n y_i x_i - \sum_{i=1}^n a x_i - \sum_{i=1}^n b x_i^2 = 0$$

(4)

$$\therefore \boxed{\sum_{i=1}^n x_i y_i = a \sum_{i=1}^n x_i + b \sum_{i=1}^n x_i^2} \text{ ---- (2)}$$

From equ (1) & equ (2) we have a system of two unknowns and we can solve them to find a & b.

Finally, substitute in $\boxed{y = a + bx}$

Ex #11 Use the method of least squares to find the straight line that best fit the following data?

x	1	2	3	4	5
y	14	27	40	55	68

$$\boxed{n=5}$$

Normal equations are $\begin{cases} a \cdot n + b \sum x = \sum y \\ a \sum x + b \sum x^2 = \sum xy \end{cases}$

x	y	x.y	x ²
1	14	14	1
2	27	54	4
3	40	120	9
4	55	220	16
5	68	340	25

$$\sum x = 15, \sum y = 204, \sum xy = 748, \sum x^2 = 55$$

$$5a + 15b = 204 \text{ (multiply by 3)}$$

$$15a + 55b = 748$$

$$(15a + 45b = 612)$$

$$15a + 55b = 748$$

$$10b = 136 \Rightarrow b = \frac{136}{10} = 13.6$$

$$\text{Now, } 5a + 15(13.6) = 204 \Rightarrow 5a = 204 - 204 \Rightarrow a = 0$$

$$\therefore y = a + bx \Rightarrow \boxed{y = 13.6x}$$

Ex #2) Certain experiment values of x and y are given 5
below

x	0	2	5	7
y	-1	5	12	20

If $y = a + bx$, find the approximation values of a & b ?

Solution

x	y	x^2	xy
0	-1	0	0
2	5	4	10
5	12	25	60
7	20	49	140

$$n = 4$$

$$\Sigma x = 14 \quad \Sigma y = 36 \quad \Sigma x^2 = 78 \quad \Sigma xy = 210$$

Normal Equations are:

$$\begin{cases} n \cdot a + b \Sigma x = \Sigma y \\ a \Sigma x + b \Sigma x^2 = \Sigma xy \end{cases}$$

~~36~~ $4a + 14b = 36$ ⁽¹⁾ (multiply by 7)
 $14a + 78b = 210$ (multiply by 2)

$$\Rightarrow \begin{pmatrix} 28a + 98b = 252 \\ 28a + 156b = 420 \end{pmatrix} -$$

$$156b - 98b = 420 - 252 \Rightarrow 58b = 168$$

$$\Rightarrow b = 2.8966$$

sub in equ(1) the value of b to get:

$$4a + 14(2.8966) = 36 \Rightarrow a = \frac{36 - 14(2.8966)}{4} = -1.1381$$

$$\therefore y = a + bx \Rightarrow y = -1.1381 + 2.8966x$$

Ex] The temperatures $T(^{\circ}\text{C})$ and lengths L (in mm) of a heated rod are given below. If $L = a + bT$. Find the best values for a & b ?

T	20	30	40	50	60	70
L	800.3	800.4	800.6	800.7	800.9	801.0

$$n = 6$$

Solution

T	L	T^2	T.L
20	800.3	400	16006
30	800.4	900	24012
40	800.6	1600	32024
50	800.7	2500	40035
60	800.9	3600	48054
70	801.0	4900	56070

$$\sum T = 270 \quad \sum L = 4803.9 \quad \sum T^2 = 13900 \quad \sum T.L = 216201$$

$$6a + 270b = 4803.9 \quad (*) \quad (\text{multiply by } 45) \Rightarrow \begin{cases} 270a + 12150b = 216175.5 \\ 270a + 13900b = 216201 \end{cases}$$

$$1750b = 25.5$$

$$b = 0.0146$$

Sub in (*) the value of b to get:

$$a = \frac{4803.9 - 270(0.0146)}{6} = \frac{4799.958}{6} = 800$$